

Answer Key & Teacher Notes

Session 1 – Dividing a Decimal by a Whole Number • Session 2 – Dividing by a Decimal

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

Dividing a Decimal by a Whole Number

TALK PROMPTS

- Why does the quotient's decimal point sit directly above the dividend's decimal point?
- What does checking by multiplying prove about your quotient?

PARTNER TASK

Quotient Check. One partner finds $50.4 \div 12$ (Problem 1f); the other checks by multiplying the quotient by 12. Switch roles with $18.9 \div 7$ (Problem 1e).

WATCH FOR

- Placing the decimal point by counting digits instead of lining it up above the dividend's point ($36.8 \div 4$ written as 0.92).
- Stopping at a remainder instead of annexing a zero and continuing ($23.4 \div 8$).
- Ignoring the decimal point entirely ($8.4 \div 4 = 21$).

SESSION 2 • TEACHER NOTES

Dividing by a Decimal

TALK PROMPTS

- What are you really multiplying by when you move both decimal points one place right?
- Why does the answer to $126 \div 9$ also answer $12.6 \div 0.9$?

PARTNER TASK

Rewrite and Check. One partner rewrites $14 \div 0.35$ (Problem 1a) with a whole-number divisor; the other solves and checks by multiplying. Switch with $37.2 \div 3.1$ (Problem 1b).

WATCH FOR

- Moving the decimal point in only one number (the divisor or the dividend), which changes the quotient by a power of 10.
- Using $\times 10$ when the divisor has two decimal places ($3.6 \div 0.12$ rewritten as $36 \div 12$).
- Expecting the quotient to be smaller than the dividend when the divisor is less than 1.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.NS.B.3	1	2	1	2	2	1	2	2	3	1	1
S1 • Form B	6.NS.B.3	1	2	1	2	2	1	2	2	3	1	1
S2 • Form A	6.NS.B.3	1	2	1	2	2	1	2	2	3	1	1
S2 • Form B	6.NS.B.3	1	2	1	2	2	1	2	2	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

dividend: The number being **divided**; the amount you share.

divisor: The number you **divide** by.

quotient: The **answer** to a division problem.

annex a zero: Write a 0 at the end of a decimal; the **value** stays the same.

- Put the decimal point in the quotient straight **above** the point in the dividend.
- If an amount is left over, **annex** a zero and keep dividing.
- Check: quotient $\times$ divisor = **dividend**.

Example: Write $25.2 \div 6$ and put the point in the quotient straight **above** the point in 25.2. 6 goes into 25 **4** times with 1 left over. Bring down the 2 to make 12. $12 \div 6 = 2$, so the quotient is 4.2. Answer: Each piece is **4.2** m. Check: $4.2 \times 6 = 25.2$

YOU TRY

- a. 2.1 b. 2.7 c. 3.75
- $\$9.60 \div 4 = \2.40

EXIT TICKET

- a. 9.3 b. 1.575 c. 3.24
- $27.6 \div 8 = 3.45$ kg (annex a zero: 27.60)
- B** (The answer is 5.25.)

PRACTICE FORM A

- a. 3.9 b. 14.1 c. 3.75 d. 3.57 e. 2.7 f. 4.2
- a. $\$96.75 \div 5 = \19.35 b. $43.2 \div 8 = 5.4$ m (check: $19.35 \times 5 = 96.75$; $5.4 \times 8 = 43.2$)
- $74.4 \div 6 = 12.4$ m
- Annex zeros (23.400) and keep dividing: $23.4 \div 8 = 2.925$ m
- $\$91.20 \div 16 = \5.70 ; Trey is not right ($5.07 \times 16 = 81.12$)
- $\$46.35 \div 5 = \9.27
- a. $58.8 \div 7 = 8.4$ m; $58.8 \div 14 = 4.2$ m b. Twice as many pieces, so each piece is half as long.
- a. $\$591.60 \div 6 = \98.60 b. $8 \times \$98.60 = \788.80 (check: $6 \times \$98.60 = \591.60)
- He put the decimal point in the wrong place. The point goes straight above the point in 36.8, and $36 \div 4 = 9$, so the answer is about 9: $36.8 \div 4 = 9.2$.
- B** (3.2)
- A** (does not change its value)

Self-check word **DIVIDE**

PRACTICE FORM B

- a. 3.15 b. 1.17 c. 3.75 d. 1.9 e. 5.28 f. 0.95
- a. $\$83.70 \div 6 = \13.95 b. $37.8 \div 4 = 9.45$ m (check: $13.95 \times 6 = 83.70$; $9.45 \times 4 = 37.8$)
- $86.4 \div 6 = 14.4$ m
- Annex zeros (30.600) and keep dividing: $30.6 \div 8 = 3.825$ m
- $\$81.60 \div 16 = \5.10 ; Ava is not right ($5.01 \times 16 = 80.16$)
- $\$58.65 \div 5 = \11.73
- a. $67.2 \div 7 = 9.6$ m; $67.2 \div 14 = 4.8$ m b. Twice as many pieces, so each piece is half as long.
- a. $\$731.40 \div 6 = \121.90 b. $9 \times \$121.90 = \$1,097.10$ (check: $6 \times \$121.90 = \731.40)
- She put the decimal point in the wrong place. The point goes straight above the point in 45.6, and $45.6 \div 6$ is about 7, so $45.6 \div 6 = 7.6$.
- C** (1.47)
- C** (straight above the point in 15.6)

Self-check word **TENTHS**

GUIDED NOTES

divisor: The number you **divide** by.

equivalent division: A new division problem with the **same** quotient.

power of ten: 10, 100, 1,000 and so on; each one is 10 times the **last**.

- Multiply the divisor and the dividend by the **same** power of 10 until the divisor is a **whole** number.
- The quotient does not **change**: $5 \div 0.25 = 500 \div 25 = 20$.

Example: Write $8.4 \div 0.7$. The divisor has **1** decimal place. Multiply both by 10: 0.7 becomes **7** and 8.4 becomes **84**. $84 \div 7 = 12$. Check: $12 \times 0.7 = 8.4$ Answer: **12** bottles

YOU TRY

1. a. $64 \div 8 = 8$ b. $27 \div 3 = 9$ c. $180 \div 6 = 30$
2. $96 \div 4 = 24$ cups

EXIT TICKET

1. a. $315 \div 35 = 9$ b. $78 \div 6 = 13$ c. $330 \div 11 = 30$
2. $156 \div 12 = 13$ labels
3. **C** (greater than 6)

PRACTICE FORM A

1. a. $1,400 \div 35 = 40$ b. $372 \div 31 = 12$ c. $126 \div 9 = 14$
d. $500 \div 25 = 20$ e. $84 \div 2 = 42$ f. $650 \div 5 = 130$
2. a. $72 \div 6 = 12$ beakers b. $45 \div 9 = 5$ tubes
3. $117 \div 13 = 9$ cans
4. $6.3 \div 7 = 0.9$ is 10 times too small. Correct: $63 \div 7 = 9$
5. $204 \div 12 = 17$ bags
6. $91 \div 7 = 13$ tails
7. She must multiply both by 100: 3.6 becomes 360, not 36. Correct: $360 \div 12 = 30$
8. a. $126 \div 6 = 21$ doors b. $126 \div 9 = 14$ doors c. The same paint is shared in bigger portions, so fewer portions fit.
9. $56 \div 8 = 7$, a whole number; Oren put a decimal point back in. The equivalent problem has the same quotient, so $5.6 \div 0.8 = 7$.
10. **B** (9)
11. **B** (100)

Self-check word DECIMAL

PRACTICE FORM B

1. a. $240 \div 15 = 16$ b. $975 \div 25 = 39$ c. $192 \div 8 = 24$ d. $225 \div 15 = 15$ e. $54 \div 9 = 6$ f. $88 \div 4 = 22$
2. a. $84 \div 7 = 12$ pots b. $56 \div 8 = 7$ trays
3. $143 \div 11 = 13$ cans
4. $7.2 \div 9 = 0.8$ is 10 times too small. Correct: $72 \div 9 = 8$
5. $306 \div 18 = 17$ bags
6. $112 \div 8 = 14$ tails
7. She must multiply both by 100: 4.2 becomes 420, not 42. Correct: $420 \div 12 = 35$
8. a. $168 \div 8 = 21$ doors b. $168 \div 12 = 14$ doors c. The same paint is shared in bigger portions, so fewer portions fit.
9. $48 \div 6 = 8$, a whole number; Malik put a decimal point back in. The equivalent problem has the same quotient, so $4.8 \div 0.6 = 8$.
10. **D** (9)
11. **C** (equivalent)

Self-check word DIVISOR

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 • DIVIDING A DECIMAL BY A WHOLE NUMBER

1. a. 3.2 b. 3.7 c. 0.42 d. 0.07 e. 5.2 f. 1.55
2. $\$16.80 \div 12 = \1.40 per pound (check: $1.40 \times 12 = 16.80$)
3. $10.2 \div 8 = 1.275$ km (annex two zeros: 10.200)
4. a. $\$3.56 \div 4 = \0.89 b. $\$5.10 \div 6 = \0.85 c. Store B (4 cents less per can)
5. $4.35 \div 5$ must be less than 1 because $4.35 < 5$. Marcus put the decimal point in the wrong place (the point goes straight above the point in 4.35). Correct quotient: 0.87.

SHEET 2 • DIVIDING BY A DECIMAL

1. a. $48 \div 6 = 8$ b. $27.5 \div 5 = 5.5$ c. $96 \div 12 = 8$ d. $90 \div 3 = 30$ e. $168 \div 4 = 42$ f. $132 \div 11 = 12$
2. $7.5 \div 1.25 = 750 \div 125 = 6$ pillows
3. a. $2.4 \div 0.3 = 24 \div 3 = 8$ glasses b. $2.4 \div 0.2 = 24 \div 2 = 12$ glasses (smaller glasses, so more of them)
4. a. $4.5 \div 0.75 = 450 \div 75 = 6$ laps b. $6 \div 0.75 = 600 \div 75 = 8$ laps
5. She moved the divisor's point two places (0.08 to 8) but the dividend's point only one place (2.4 to 24). Both must move two places: $240 \div 8 = 30$.
6. **B** ($67.2 \div 8$)
7. **C** (39)